MAT 3007 Homework # 7 Spring 2024

Due: Monday, May 6 at 12:00pm (noon)

Please note: You should submit your answers (scanned or typeset) in a PDF file. Question 2 and 3 asks you to write codes in Python or MATLAB. Submit all necessary code files via Blackboard. Also, give meaningful names to your decision variables and parameters, and add comments to your code liberally. The problem set is due at 12:00pm (noon) on Monday, May 6th.

1. (20 pts) Perform one iterations of gradient descent method for the following problem with the exact line search to find the step length:

$$\min \quad 3x_1^2 + 4x_2^2 - x_1x_2 - 5x_1 - 8x_2$$

2. (20 pts) We consider the optimization problem:

$$\begin{aligned} \min_{x} \quad & -\frac{1}{(x-1)^2}[\log(x) - \frac{2(x-1)}{x+1}] \\ \text{s.t.} \quad & 1.5 \leq x \leq 4.5. \end{aligned}$$

Implement the bisection **or** the golden section method (using MATLAB or Python) to solve this problem and output a solution with accuracy at least 10^{-5}.

3. (30 pts) Implement the gradient descent method to solve the following problem:

$$\min_{x\in\mathbb{R}^2} \quad (3 + x_1 + ((1 - x_2)x_2 - 2)x_2)^2 + (3 + x_1 + (x_2 - 3)x_2)^2.$$

The following input functions and parameters should be considered:

- `obj`, `grad` – function handles that calculate and return the objective function $f(x)$ and the gradient $\nabla f(x)$ at an input vector $x \in \mathbb{R}^n$. You can treat these handles as functions or fields of a class or structure `f` or use them directly as input. (For example, your function can have the form `gradient_method(obj,grad,...)`).
- x^0 – the initial point.
- `tol` – a tolerance parameter. The method should stop whenever the current iterate x^k satisfies the criterion $\|\nabla f(x^k)\| \leq$ `tol`.

We want to analyze the performance of the gradient method for different step size strategies. In particular, we want to test and compare backtracking, exact line search, and diminishing step sizes. The following parameters will be relevant for these strategies:

- $\beta, \gamma \in (0, 1)$ – parameters for backtracking and the Armijo condition.
- You can use the golden section method to determine the exact step size α_k. The parameters for the golden section method are: `maxit` (maximum number of iterations), $\texttt{tol}_\alpha$ (stopping tolerance), $[0, \texttt{a}]$ (the interval of the step size).

You can organize the latter parameters in an appropriate `options` class or structure. It is also possible to implement two separate algorithms for backtracking and diminishing step sizes. The method(s) should return the final iterate x^k that satisfies the stopping criterion.

(a) Verify that all stationary points of the mapping f are given by:

$$x_1^* = \begin{pmatrix} -3 \\ 0 \end{pmatrix}, \quad x_2^* = \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \quad x_2^* = \begin{pmatrix} -7 \\ -1 \end{pmatrix}, \quad x_4^* = \begin{pmatrix} -\frac{10}{3} + \frac{8\sqrt{3}}{9} \\ \frac{\sqrt{3}}{3} \end{pmatrix}, \quad x_5^* = \begin{pmatrix} -\frac{10}{3} - \frac{8\sqrt{3}}{9} \\ -\frac{\sqrt{3}}{3} \end{pmatrix}.$$

(b) Which of the stationary points are (global/local) minimizer, (global/local) maximizer, or saddle points?

(c) Apply the gradient method with backtracking and parameters $(\beta, \gamma) = (0.5, 0.1)$ and exact line search ($\texttt{maxit} = 100$, $\texttt{tol}_\alpha = 10^{-6}$, $\texttt{a} = 2$) to solve the problem $\min_x f(x)$.

The algorithms should use the stopping tolerance $\texttt{tol} = 10^{-5}$. Test the methods using the initial point $(0, 0)^\top$ and report the behavior and performance of the methods. In particular, for each of the initial points, compare the number of iterations and the point to which each algorithm converged.

4. (30 pts) Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuously differentiable, $x^k \in \mathbb{R}^n$, and let d^k be a descent direction of f at x^k. We again consider the function $\phi(\alpha) := f(x^k + \alpha d^k) - f(x^k)$, $\alpha \geq 0$.

For a given parameter $\gamma \in (0, 1)$, a step size $\alpha \geq 0$ is said to satisfy the Armijo condition if $\phi(\alpha) \leq \gamma\alpha \cdot \phi'(0)$. Backtracking or the Armijo line search technique tries to find the largest step size $\alpha \in \{1, \beta, \beta^2, \beta^3, ...\}$, $\beta \in (0, 1)$, that satisfies the Armijo condition. Exact line search determines the step size $\bar{\alpha}$ as global minimizer of ϕ: $\bar{\alpha} = \arg\min_{\alpha \geq 0} \phi(\alpha)$.

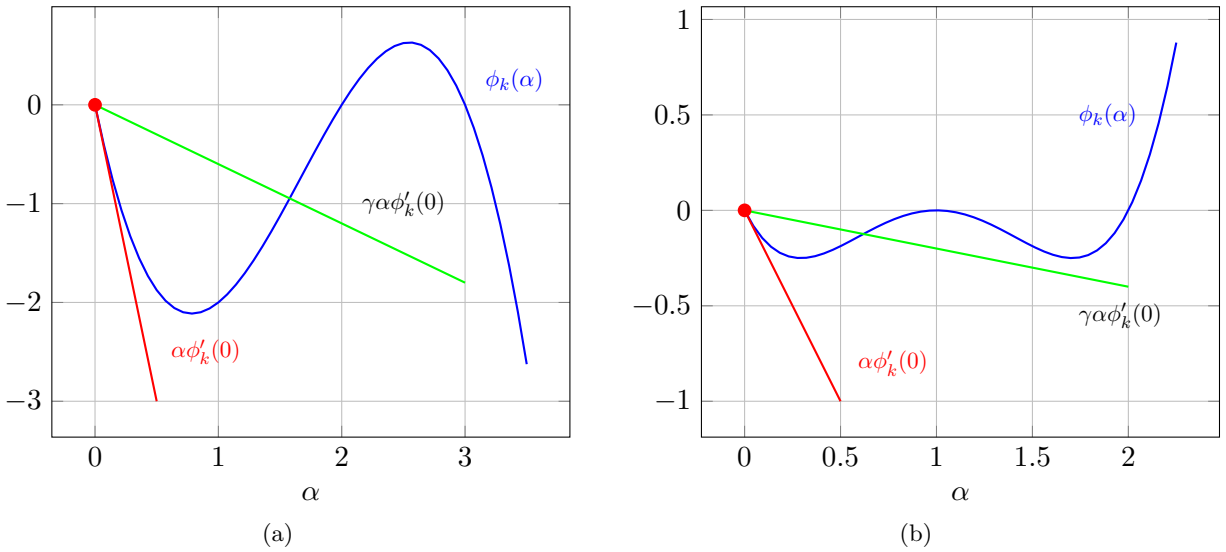


Figure 1: Multiple choice & step sizes

(a) Consider Figure 1(a). For each of the following, indicate whether the statement is true and explain:

i) Backtracking returns the step size $\alpha = 1$.

ii) The Armijo condition is satisfied for all $\alpha \in [0, 1]$.

iii) The exact step size $\bar{\alpha}$ lies in the interval $[0.5, 1]$.

(b) Consider Figure 1(b). For each of the following, indicate whether the statement is true and explain:

i) Suppose that $\beta = \frac{1}{2}$. Then, backtracking returns the step size $\alpha = \frac{1}{2}$.

ii) If $\phi(\alpha) \to \infty$ for $\alpha \to \infty$, then the exact step size $\bar{\alpha}$ lies in the interval $[0, 0.5]$.

iii) If $\beta \leq \frac{1}{2}$, then Armijo line search returns the step size $\alpha = \beta$.